
GIT PUSH THE BEATS

Melody Dive **Use-Case Specification: Manage music catalog**

Version 1.0

Melody Dive	Version: 1.0
Use-Case Specification: Manage music catalog	Date: 06/12/2025
UC Manage music catalog	

Revision History

Date	Version	Description	Author
15/11/2025	1.0	The initial version	Huỳnh Gia Bảo
06/12/2025	2.0	Change prototype UI to dark mode	Huỳnh Gia Bảo

Melody Dive	Version: 1.0
Use-Case Specification: Manage music catalog	Date: 06/12/2025
UC Manage music catalog	

Table of Contents

1. Use-Case Name: Manage music catalog	4
1.1 Brief Description	4
2. Flow of Events	4
2.1 Basic Flow	4
3. Manage songs	4
2.2 Alternative Flows	9
2.2.1 Existed song/playlist	9
3. Special Requirements	9
3.1 Speed	9
3.2 Concurrence	9
4. Preconditions	10
4.1 Admin account logged in	10
5. Postconditions	10
5.1 Song/playlist updated	10

Melody Dive	Version: 1.0
Use-Case Specification: Manage music catalog	Date: 06/12/2025
UC Manage music catalog	

Use-Case Specification: Manage music catalog

1. Use-Case Name: Manage music catalog

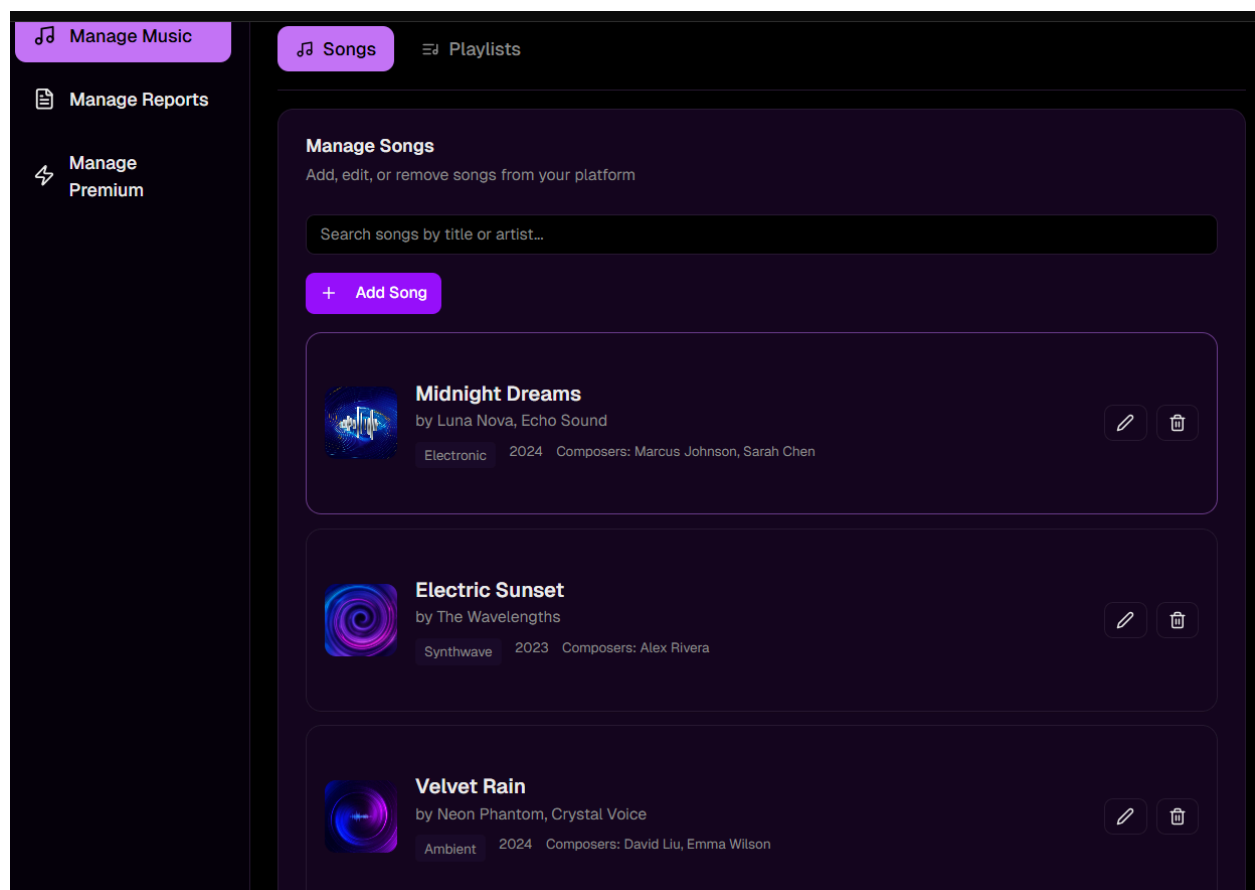
1.1 Brief Description

This use-case describes how admins can add, remove, and modify songs and playlists of songs.

2. Flow of Events

2.1 Basic Flow

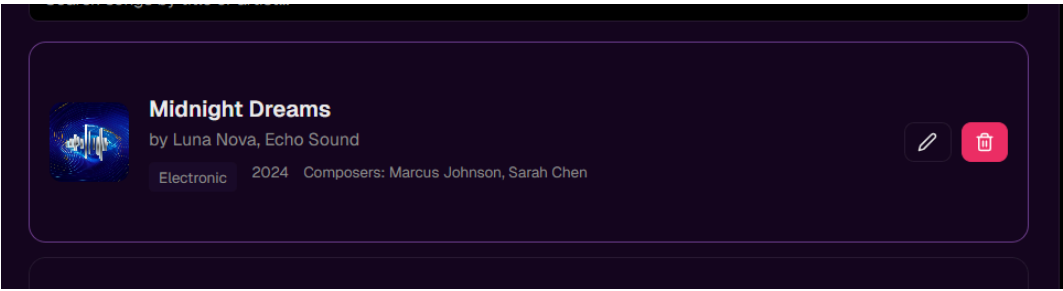
1. Admin selects *Manage music catalog* section.
2. Admin can see 2 sections: *Songs* and *Playlists*, choose one section to view all songs/playlists.



3. Manage songs

- Remove a song:
 - + Admin selects a song to view options
 - + Then, admin chooses **Delete song**.
 - + Then, the system deletes the song from the platform.

Melody Dive	Version: 1.0
Use-Case Specification: Manage music catalog	Date: 06/12/2025
UC Manage music catalog	



- Add a song:
 - + Admin selects **Add song** to add song's audio file, its avatar picture and its metadata (name, author's name, composer's name, genre, published year, lyrics, etc).
 - + If the song already exists, it goes to the *Existed song* alternative flow mentioned below, or else the system will receive the data and create the song according to those data.

A screenshot of a "Add New Song" form. The form is titled "Add New Song" with a close button (X) in the top right corner. It contains several input fields: "Song Title", "Genre", "Artists (comma-separated)", "Published Year", "Composers (comma-separated)", "Audio File", "Lyrics File", and "Avatar Image". Each of the last three fields has a "Choose File" button and a "No file chosen" status. At the bottom right, there are "Cancel" and "Add Song" buttons.

- Modify a song:
 - + Admin selects a song to view options.
 - + Admin then chooses **Modify song** to modify its metadata, or to change its avatar picture.
 - + The system updates the song.

Melody Dive	Version: 1.0
Use-Case Specification: Manage music catalog	Date: 06/12/2025
UC Manage music catalog	

✕

Edit Song

Song Title

Midnight Dreams

Genre

Electronic

Artists (comma-separated)

Luna Nova, Echo Sound

Published Year

2024

Composers (comma-separated)

Marcus Johnson, Sarah Chen

Audio File

Choose File

No file chosen

✓ midnight-dreams.mp3

Lyrics File

Choose File

No file chosen

Avatar Image

Choose File

No file chosen

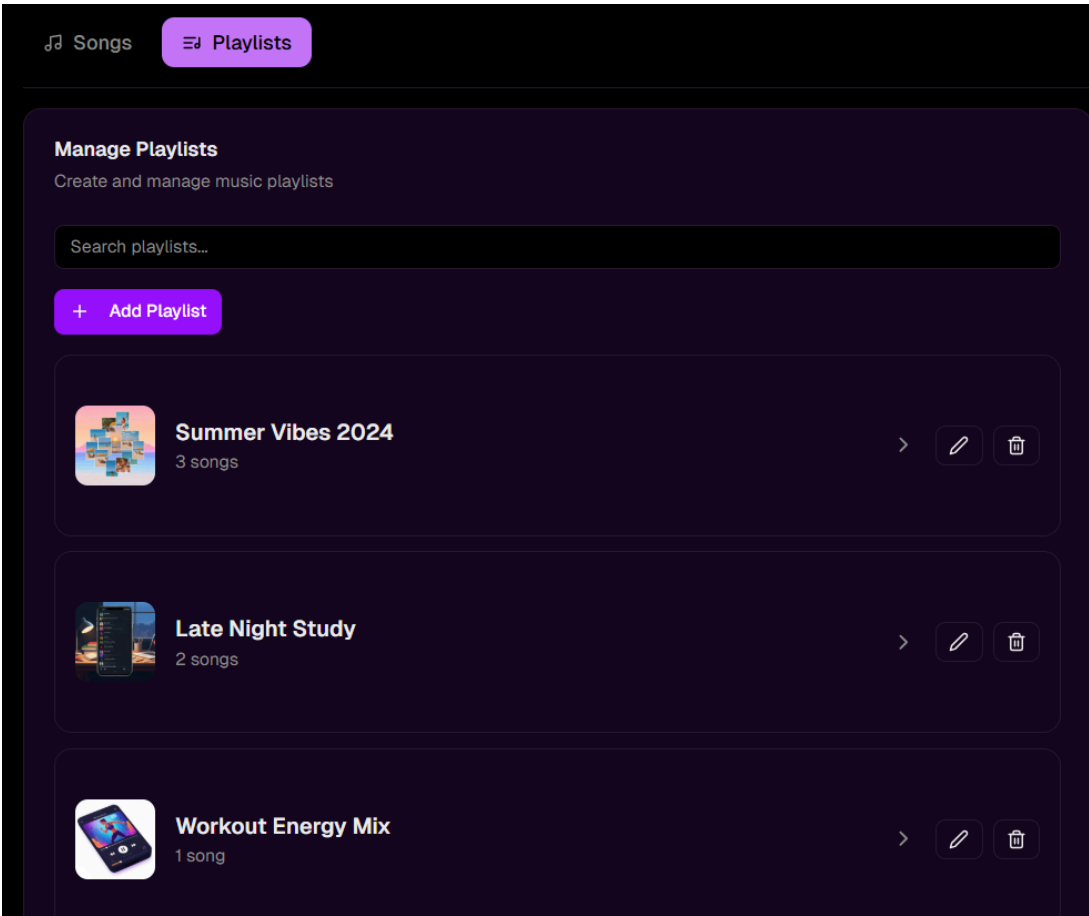
✓ /album-cover-1.png

Cancel

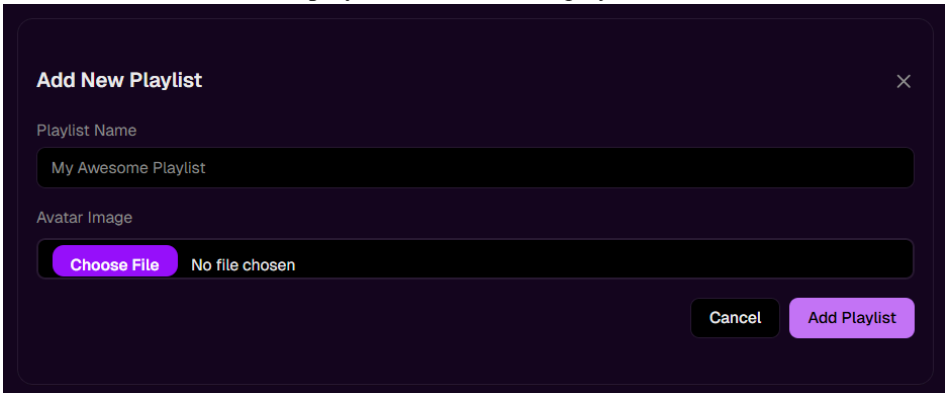
Update Song

4. *Manage playlists*

Melody Dive	Version: 1.0
Use-Case Specification: Manage music catalog	Date: 06/12/2025
UC Manage music catalog	



- Remove a playlist:
 - + Admin selects a playlist to view options.
 - + Admin then chooses **Delete playlist**.
 - + The system deletes the playlist.
- Add a playlist:
 - + Admin selects **Add playlist** to create a new playlist.



- + Admin then names it and uploads its avatar picture.
- + The system checks if the playlist's name already exists. If not, the playlist is generated.

Melody Dive	Version: 1.0
Use-Case Specification: Manage music catalog	Date: 06/12/2025
UC Manage music catalog	

- Modify a playlist:
 - + Admin selects a playlist to view options.
 - + Admin then chooses **Modify playlist** to modify it. Specifically, admin can change its name, and avatar picture.
 - + The system checks if the playlist's name already exists. If not, the playlist is updated.

Edit Playlist ✕

Playlist Name

Summer Vibes 2024

Avatar Image

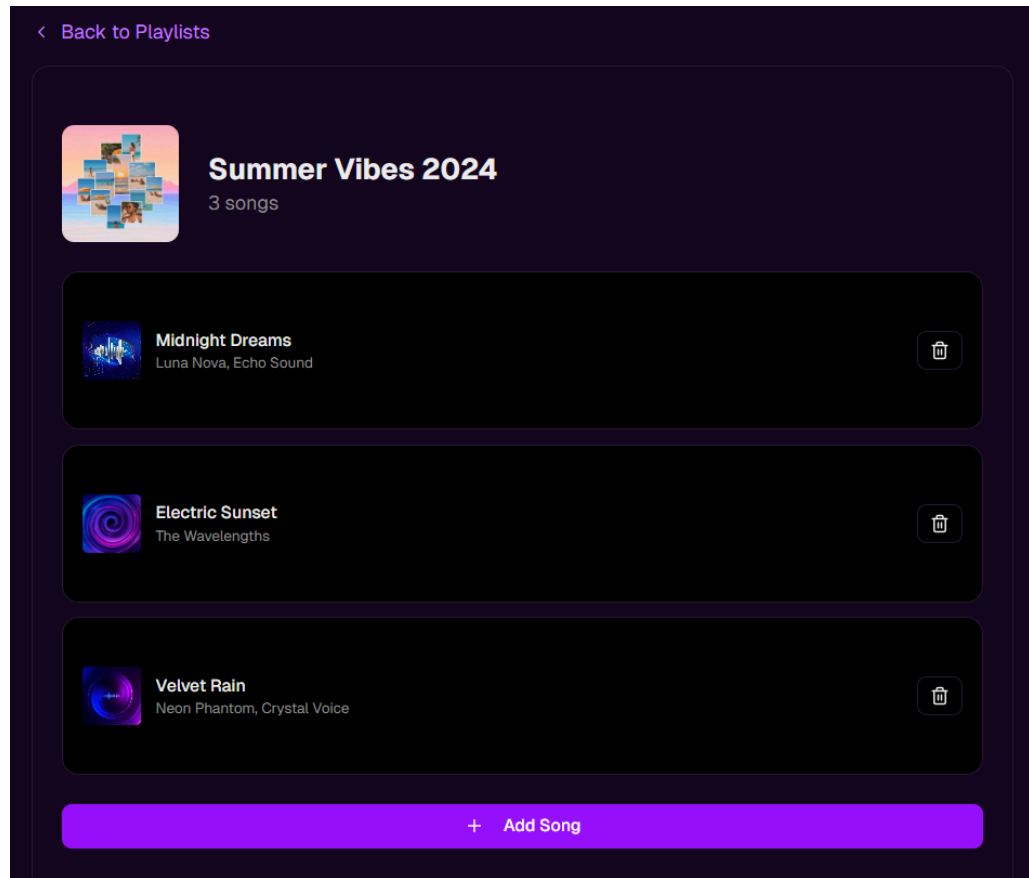
Choose File No file chosen

✓ /summer-playlist.jpg

Cancel Update Playlist

- Manage songs in a playlist:
 - + Admin goes to a specific playlist to view all its songs
 - + Then, admin follows the steps mentioned above in **Manage song**, the flow will be the same as managing song.

Melody Dive	Version: 1.0
Use-Case Specification: Manage music catalog	Date: 06/12/2025
UC Manage music catalog	



2.2 Alternative Flows

2.2.1 Existed song/playlist

This alternative flow starts when the admin adds a song/playlist which already existed, or when the admin modifies a song/playlist then makes it become an existed one.

A song is defined 'existed' if its metadata already exists., and a playlist is defined 'existed' if its name already exists.

In this flow, an error message will be displayed and the song/playlist will not be created/modified.

3. Special Requirements

3.1 Speed

The time to upload a song must be less than a quarter of the song's duration. (eg: Uploading a song lasting 4 minutes must take less than 1 minute.)

Metadata takes less than 1 second to be updated.

3.2 Concurrency

The system allows up to 10 admins to upload songs at the same time, and still ensures the speed mentioned above.

Melody Dive	Version: 1.0
Use-Case Specification: Manage music catalog	Date: 06/12/2025
UC Manage music catalog	

4. Preconditions

4.1 Admin account logged in

The admin must have logged in an **Admin** account.

5. Postconditions

5.1 Song/playlist updated

The song/playlist is added/removed/modified.